វិទ្យាស្ថានបច្ចេកវិទ្យាកំពង់ស្ពឺ

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ការរៀបចំបច្ច័យពិសោធន៍ដែលមាន 3 factors, 2 levels នៅក្នុង CRD

មុខវិជ្ជា ៖ការរៀបចំផែនការពិសោធន៍

បង្រៀនដោយលោកគ្រូ ៖ ម៉ម សំឡី

ដេប៉ាតឺម៉ង់ ៖ វិស្វកម្មគីមីចំណីអាហារ

ជំនាញ ៖ បច្ចេកវិទ្យាអាហារ

សេង ធានី

ឆ្នាំសិក្សា ២០២២ - ២០២៣

3 factors, 2 levels

- Factor A = Temperature 2 levels : a_1, a_2
- Factor B = Concentration 2 levels : b_1, b_2
- Factor C = Time 2 levels : c_1, c_2

  - 2 x 2 x 2 factorial

  2^3 factorial

Preparation experimental

A: Temp	B: Conc.	C: Time
a_1	b_1	c_1
a_1	b_1	c_2
a_1	b_2	c_1
a_1	b_2	c_2
a_2	b_1	c_1
a_2	b_1	c_2
a_2	b_2	c_1
a_2	b_2	c_2

Treatment
$a_1b_1c_1$
$a_1b_1c_2$
$a_1b_2c_1$
$a_1b_2c_2$
$a_2b_1c_1$
$a_2b_1c_2$
$a_2b_2c_1$
$a_2b_2c_2$

ANOVA calculation for 3 factorial in CRD

SOV	DF	SS	MS	F
Treatment	t-1	SST	MST	MST / MSE
A	a-1	SSA	MSA	MSA / MSE
B	b-1	SSB	MSB	MSB / MSE
C	c-1	SSC	MSC	MSC / MSE
A x B	(a-1)(b-1)	SS(AB)	MS(AB)	MS(AB) / MSE
A x C	(a-1)(c-1)	SS(AC)	MS(AC)	MS(AC) / MSE
B x C	(b-1)(c-1)	SS(BC)	MS(BC)	MS(BC) / MSE
A x B x C	(a-1)(b-1)(c-1)	SS(ABC)	MS(ABC)	MS(ABC) / MSE
Error	t(r-1)	SSE	MSE	
Total	tr-1	Total SS		

r = ចំនួនដងដែលៗ

t = ចំនួនទាំងអស់របស់បច្ច័យពិសោធន៍= abc

a = ចំនួនសរុបរបស់បច្ច័យA

b = ចំនួនសរុបរបស់បច្ច័យB

c = ចំនួនសរុបរបស់បច្ច័យC

R_r = ផលបូកនៃការធ្វើម្តងទៀតនៃ r នីមួយៗ

$T_t = t$

A_a = sum of each level of A

B_b = sum of each level of B

C_c = sum of each level of C

$A_a B_b$ = The sum of each combination of AB

$A_a C_c$ = The sum of each combination of AC

$B_b C_c$ = The sum of each combination of BC

$SST = (T_1^2 + T_2^2)/r - CF$

$SSA = (a_1^2 + a_2^2)/bcr - CF$

$SSB = (b_1^2 + b_2^2)/acr - CF$

$SSC = (c_1^2 + c_2^2)/abr - CF$

$SS(AB) = (a_1b_1^2 + a_1b_2^2 + a_2b_1^2 + a_2b_2^2)/cr - CF - SSA - SSB$

$SS(AC) = (a_1c_1^2 + a_1c_2^2 + a_2c_1^2 + a_2c_2^2)/br - CF - SSA - SSC$

$SS(BC) = (b_1c_1^2 + b_1c_2^2 + b_2c_1^2 + b_2c_2^2)/ar - CF - SSB - SSC$

$SS(ABC) = SST - SSA - SSB - SSC - SS(AB) - SS(AC) - SS(BC)$

$SSE = \text{Total SS} - SST$

$\text{Total SS} = \sum(\text{each value}^2) - CF$

$CF = (\sum(Y_{ij}))^2$ / ចំនួនទិន្នន័យសរុប(Total amount of data)

$$MST = SST/(t-1)$$

$$MSA = SSA/(a-1)$$

$$MSB = SSB/(b-1)$$

$$MSC = SSC/(c-1)$$

$$MS(AB) = SS(AB)/(a-1)(b-1)$$

$$MS(AC) = SS(AC)/(a-1)(c-1)$$

$$MS(BC) = SS(BC)/(b-1)(c-1)$$

$$MS(ABC) = SS(ABC)/(a-1)(b-1)(c-1)$$

$$MSE = SSE/t(r-1)$$

EXAMPLE

want to develop production methods, plan experiments factorial in CRD by studying 3 factors with 2 levels of each factor, namely

Treatment A = temperature 100 and 120 oC

(a_1 = 100 oC, a_2 = 120 oC)

Treatment B = Time 60mn and 80mn

(b_1 = 60mn, b_2 = 80mn)

Treatment C = Sample preparation is old and new.

(c_1 = old, c_2 = new)

The experimental sample was collected 3 times and then used to meas ure the quality. The experimenter wanted to know whether the factors of production influenced the quality value or not. And there is a relatio nship between these factors or not.

Table of experimental results

(a1 = 100, a2 = 120)

(b1 = 60, b2 = 80)

(C1 = 1, c2 = 2)

Treatment	Rep 1	Rep 2	Rep 3
$a_1b_1c_1$	22	31	25
$a_1b_1c_2$	44	45	38
$a_1b_2c_1$	35	34	50
$a_1b_2c_2$	60	50	54
$a_2b_1c_1$	32	43	29
$a_2b_1c_2$	40	37	36
$a_2b_2c_1$	55	47	46
$a_2b_2c_2$	39	41	47

From the problem, let me know

- r = ចំនួនដដែលៗ = 3
- a = ចំនួនសរុបនៃបច្ច័យ A = 2
- b = ចំនួនសរុបនៃបច្ច័យ B = 2
- c = ចំនួនសរុបនៃបច្ច័យ C = 2
- t = ចំនួនសរុបនៃការពិសោធន៍របស់បច្ច័យទាំងអស់ = abc = 2x2x2 = 8

១. ការកំណត់សម្មតិកម្មពិសោធន៍

- Ho : $\mu_{a1} = \mu_{a2}$
- Ho : $\mu_{b1} = \mu_{b2}$
- Ho : $\mu_{c1} = \mu_{c2}$
- Ho : interaction between factors(អន្តរកម្មរវាងកត្តា A, B និង C = 0 ឬមិនមែន interaction between A, B និងC

๒. Calculation

Treatment	ซ้ำ1 (repetitive'	ซ้ำ 2	ซ้ำ 3
$a_1b_1c_1$	22	31	25
$a_1b_1c_2$	44	45	38
$a_1b_2c_1$	35	34	50
$a_1b_2c_2$	60	50	54
$a_2b_1c_1$	32	43	29
$a_2b_1c_2$	40	37	36
$a_2b_2c_1$	55	47	46
$a_2b_2c_2$	39	41	47

២.១ ផលបូកនៃបច្ច័យនីមួយៗ

Treatment	ដង 1	ដង 2	ដង 3	ΣT
$a_1b_1c_1$	22	31	25	78
$a_1b_1c_2$	44	45	38	127
$a_1b_2c_1$	35	34	50	119
$a_1b_2c_2$	60	50	54	164
$a_2b_1c_1$	32	43	29	104
$a_2b_1c_2$	40	37	36	113
$a_2b_2c_1$	55	47	46	148
$a_2b_2c_2$	39	41	47	127

២.២ ផលបូកនៃបច្ច័យ a

Treatment	ឆ្ងំា 1	ឆ្ងំា 2	ឆ្ងំា 3	ΣT
$a_1b_1c_1$	22	31	25	78
$a_1b_1c_2$	44	45	38	127
$a_1b_2c_1$	35	34	50	119
$a_1b_2c_2$	60	50	54	164
$a_2b_1c_1$	32	43	29	104
$a_2b_1c_2$	40	37	36	113
$a_2b_2c_1$	55	47	46	148
$a_2b_2c_2$	39	41	47	127

$\Sigma a1 = 488$

$\Sigma a2 = 492$

២.ពាតម្លៃផលបូកនៃបច្ច័យ b

Treatment	ซ้ำ 1	ซ้ำ 2	ซ้ำ 3	ΣT
$a_1b_1c_1$	22	31	25	78
$a_1b_1c_2$	44	45	38	127
$a_1b_2c_1$	35	34	50	119
$a_1b_2c_2$	60	50	54	164
$a_2b_1c_1$	32	43	29	104
$a_2b_1c_2$	40	37	36	113
$a_2b_2c_1$	55	47	46	148
$a_2b_2c_2$	39	41	47	127

$\boxed{\Sigma b1 = 422}$

$\Sigma b2 = 558$

Treatment	ซ้ำ 1	ซ้ำ 2	ซ้ำ 3	ΣT
$a_1b_1c_1$	22	31	25	78
$a_1b_1c_2$	44	45	38	127
$a_1b_2c_1$	35	34	50	119
$a_1b_2c_2$	60	50	54	164
$a_2b_1c_1$	32	43	29	104
$a_2b_1c_2$	40	37	36	113
$a_2b_2c_1$	55	47	46	148
$a_2b_2c_2$	39	41	47	127

$\Sigma c1 = 449$

$\Sigma c2 = 558$

២.៥ តម្លៃផលបូកនៃបច្ច័យ ab

Treatment	ចំា 1	ចំា 2	ចំា 3	ΣT	
$a_1b_1c_1$	22	31	25	78	Σa1b1 = 205
$a_1b_1c_2$	44	45	38	127	
$a_1b_2c_1$	35	34	50	119	Σa1b2 = 283
$a_1b_2c_2$	60	50	54	164	
$a_2b_1c_1$	32	43	29	104	Σa2b1 = 217
$a_2b_1c_2$	40	37	36	113	
$a_2b_2c_1$	55	47	46	148	Σa2b2 = 275
$a_2b_2c_2$	39	41	47	127	

២.៦ តម្លៃផលបូកនៃបច្ច័យ ac

Treatment	ដង 1	ដង 2	ដង 3	ΣT
$a_1b_1c_1$	22	31	25	78
$a_1b_1c_2$	44	45	38	127
$a_1b_2c_1$	35	34	50	119
$a_1b_2c_2$	60	50	54	164
$a_2b_1c_1$	32	43	29	104
$a_2b_1c_2$	40	37	36	113
$a_2b_2c_1$	55	47	46	148
$a_2b_2c_2$	39	41	47	127

$\Sigma a1c1 = 197$

$\Sigma a1c2 = 291$

$\Sigma a2c1 = 252$

$\Sigma a2c2 = 240$

២.ពតម្លៃផលបូកនៃបច្ច័យ bc

Treatment	ซ้ำ 1	ซ้ำ 2	ซ้ำ 3	ΣT
$a_1b_1c_1$	22	31	25	78
$a_1b_1c_2$	44	45	38	127
$a_1b_2c_1$	35	34	50	119
$a_1b_2c_2$	60	50	54	164
$a_2b_1c_1$	32	43	29	104
$a_2b_1c_2$	40	37	36	113
$a_2b_2c_1$	55	47	46	148
$a_2b_2c_2$	39	41	47	127

Σb1c1 = 182

Σb1c2 = 240

Σb2c1 = 267

Σb2c2 = 291

២.៨ គណនាតម្លៃតាមតារាងANOVA

- $CF = (\Sigma(Y_{ij}))^2$ / total number of data

  $= (\text{Total})^2 / (rabc)$

  $= (980)^2 / (3x2x2x2)$

  $= 40016.67$

- Total SS $= \Sigma(\text{each value})^2 - CF$

  $= 42112 - 40016.67$

  $= 2095.33$

- SST $= (T_1^2 + T_2^2 + \ldots + T_t^2)/r - CF$

  $= (78^2 + 127^2 + 119^2 + \ldots + 127^2)/3 - CF$

  $= 1612.67$

- SSE $=$ Total SS $-$ SST $= 482.67$

- SSA $= (a_1^2 + a_2^2)/bcr - CF$

  $= (488^2 + 492^2)/(2x2x3) - CF = 0.66$

- SSB $= (b_1^2 + b_2^2)/acr - CF$

  $= (422^2 + 558^2)/(2x2x3) - CF = 770.66$

- SSC $= (c_1^2 + c_2^2)/abr - CF$

  $= (449^2 + 531^2)/(2x2x3) - CF = 280.16$

- SS(AB) $= (a_1b_1^2 + a_1b_2^2 + a_2b_1^2 + a_2b_2^2)/cr - CF - SSA - SSB$

  $= 16.68$

- SS(AC) $= (a_1c_1^2 + a_1c_2^2 + a_2c_1^2 + a_2c_2^2)/br - CF - SSA - SSC$

  $= 468.18$

- SS(BC) $= (b_1c_1^2 + b_1c_2^2 + b_2c_1^2 + b_2c_2^2)/ar - CF - SSB - SSC$

  $= 48.18$

- SS(ABC)

  $= SST - SSA - SSB - SSC - SS(AB) - SS(AC) - SS(BC)$

  $= 28.15$

- MST = SST/(t-1) = 230.38
- MSA = SSA/(a-1) = 0.66
- MSB = SSB/(b-1) = 770.66
- MSC = SSC/(c-1) = 280.16
- MS(AB) = SS(AB)/(a-1)(b-1) = 16.68
- MS(AC) = SS(AC)/(a-1)(c-1) = 468.18
- MS(BC) = SS(BC)/(b-1)(c-1) = 48.18
- MS(ABC) = SS(ABC)/(a-1)(b-1)(c-1) = 28.15
- MSE = SSE/(t-1)(r-1) = 30.17

SOV	DF	SS	MS	F	F table ที่ α = 0.05
Trt	7	1612.67	230.38	6.68*	2.77
A	1	0.66	0.66	0.02	4.60
B	1	770.66	770.66	22.38*	4.60
C	1	280.16	280.16	8.14*	4.60
AB	1	16.68	16.68	0.48	4.60
AC	1	468.18	468.18	13.60*	4.60
BC	1	48.18	48.18	1.40	4.60
ABC	1	28.15	28.15	0.82	4.60
Error	16	482.67	30.17		
Total	23	2095.33			

៣. Conclusion

Summary of experimental results

- Have an average of at least 1 pair that differs(មានជាមធ្យមយ៉ាងហោចណាស់ 1 គូដែលខុសគ្នា)
- Factors B and C influence product quality values.(កត្តា B និង C មានឥទ្ធិពលលើតម្លៃគុណភាពផលិតផល។)
- There is a correlation between factor A and C in an experiment at a 95% confidence level.(មានការជាប់ទាក់ទងគ្នារវាងកត្តា A និង C ក្នុងការពិសោធន៍នៅកម្រិតទំនុកចិត្ត 95%)

exercise

From the 3x3x2 Factorial in CRD experiment that was performed 3 times, the factors were as follows:(ការពិសោធន៍ 3x3x2 Factorial ក្នុងការពិសោធន៍ CRD ដែលត្រូវបានអនុវត្ត 3 ដង ៖)

- Factor A is temperature, there are 3 levels.
- Factor B is speed, there are 2 levels.
- Factor C is pressure, there are 2 levels.

ចង់សិក្សាពីផលប៉ះពាល់នៃកត្តាទាំង៣នេះថាតើវានឹងប៉ះពាល់ដល់ viscosity របស់ផលិតផលបង្អែមឬអត់? ហើយមានទំនាក់ទំនងរវាងកត្តាឬអត់។ តម្លៃ viscosity ត្រូវបានបង្ហាញក្នុងតារាង។

1. Make an experimental hypothesis(សម្មតិកម្ម)
2. វិភាគរកភាពខុសគ្នានៃការពិសោធន៍នៅកម្រិតជឿជាក់ 0.05
3. Summarize the experimental results.

The viscosity values are shown in the table.

	b1		b2	
	c1	c2	c1	c2
a1	32 31 36	44 45 45	35 34 33	60 55 52
a2	42 43 40	40 37 36	55 47 50	39 41 45
a3	25 29 27	38 36 36	50 46 51	54 47 49


THANK
YOU!